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Physical Modeling of Shallow Karst Systems Using Additive Manufacturing Techniques: Applications in Drilling Hazard Evaluation and Prevention

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Abstract

Karstification is a widespread phenomenon in the arid environment of the Arabian Peninsula, posing significant risks to drilling operations due to its unpredictable nature. As a result, identifying karst features through geophysical methods is essential to mitigate potential hazards. Numerical modeling has been employed to test ideas, but it often oversimplifies or neglects crucial phenomena such as visco-elasticity and attenuation, limiting its accuracy. To address this limitation, physical modeling can bridge the gap by simulating real-world complexities. This paper presents a novel approach to constructing realistic physical models that mimic the characteristics of shallow karsts, and provides an overview of the laboratory setup designed to validate methods for identifying these features.

Introduction

Near-surface voids and sinkholes resulting from the karstification process pose significant hazards to engineering, construction, and drilling activities. These geological features can disrupt drilling operations by causing loss of circulation, leading to rig instability and, in severe cases, necessitating the abandonment of wells. The accurate prediction and mapping of sinkhole-prone areas are therefore vital to mitigate these risks. Historically, various geophysical methods have been employed to detect subsurface cavities, including electrical resistivity tomography, ground-penetrating radar, microgravity, and seismic methods (Putiška et al., 2014; Gizzi et al., 2010; Martinez-Pagan et al., 2013; McNeice et al., 2017; Colombo et al., 2024).

Previous studies have demonstrated the feasibility of detecting air or liquid-filled cavities in the near-surface using passive seismic data and frequency spectra examination. Caves and underground strata can act as resonators, creating distinct resonance frequencies that distinguish them from the surrounding hard rocks. Theories explaining these resonant frequencies include trapped waves within cavities, circumferential waves, and standing waves formed between the cavity and the surface, as discussed in research by Korneev (2009), Gritto (2021), Fedin et al. (2021), and Silvestrov et al. (2024). The simplicity and independence from active sources of this method have also made it suitable for use in shallow karsting identification using autonomous UAV swarms (Golikov et al., 2024).

One approach to enhancing the precision of seismic mapping using passive records obtained with small-channel equipment is to utilize a technique that entails recalculating data obtained at different time periods to a single time using synchronous records (Emanov et al., 2002, Emanov et al., 2008). This method aids in identifying standing waves more effectively from microseismic records. It has been tested in both laboratory settings (Kolesnikov and Fedin, 2018) and in the field (Fedin et al., 2021) to identify the shape of karst caves.

Our laboratory experiments have demonstrated the significance of physical modeling and data acquisition in a controlled laboratory environment for tuning acquisition and data processing parameters. However, in our previous studies, we used an oversimplified karst model represented by a single large cylindrical void (Silvestrov et al., 2024). In this study, we demonstrate a geology-driven approach for building realistic digital karst models and their subsequent implementation in laboratory mockups using additive manufacturing techniques for designing complex surfaces of cave ceilings and floors.

The development of realistic digital karst models is crucial for accurately simulating the seismic response of karstified areas. By incorporating geological data and constraints, we can create models that mimic the complexity of real-world karst systems. The use of additive manufacturing techniques allows for the creation of intricate cave geometries, enabling the simulation of seismic wave propagation through complex karst structures. Our laboratory measurement setup is designed to simulate karst detection using the standing waves resonance method, which involves recording passive seismic data and analyzing the frequency spectra to identify distinct resonance frequencies indicative of subsurface cavities.

The experimental setup consists of a physical model of a karstified area, created using a combination of 3D printing and concrete matrix embedding. The model is then subjected to seismic testing using piezoelectric sources and ultrasonic sensor arrays. The recorded data are then processed using the technique of recalculating data obtained at different time periods to a single time using synchronous records. This approach enables the identification of standing waves and the detection of subsurface cavities. Our study aims to improve the accuracy of karst detection, ultimately reducing the risks associated with drilling and construction activities in karst-prone areas. By combining physical modeling, additive manufacturing, and advanced seismic data processing techniques, we can develop a more effective approach to identifying and characterizing near-surface voids and sinkholes.

Model building

A scaled digital 3D model was developed to represent a 500x500x500m rock volume, which was then condensed to a 0.5x0.5x0.5m physical prototype at a scale of 1:1000. The digital model consisted of a gridded structure comprising four distinct layers: a superficial top sand layer, two carbonate strata (upper and lower) with subtle variations in vertical properties, and a karstified zone situated at the interface between the two carbonate layers. The subsequent stages involved 3D printing these geological horizons and embedding them within a concrete matrix that exhibited analogous physical characteristics to the simulated formations. Following an incubation period to allow complete settling, seismic testing was initiated using piezoelectric sources and ultrasonic sensor arrays to simulate seismic wave propagation through the model. This experimental setup enabled the investigation of seismic responses in a controlled environment, mimicking real-world geological conditions.

Geological analogues

This study employed two distinct shallow karst system as analogs to simulate both simple and complex karstification system embedded within a limestone host rock. The simple analog replicates a fracture-dominated karst network, typical of karst system in Arabia (Amin and Bankher, 1997; Forti et al., 2003). This model features relatively linear conduit geometries formed primarily by the dissolutional widening of interconnected fracture sets within carbonate rock (Figure 1A). The complex analog represents a mature epigenic cave system inspired by the intricate three-dimensional morphology of

Watson Cave, Oregon (Figure 1B). The detailed geometry for these karsts were derived by digitally processing comprehensive spelunker survey data (https://npshistory.com/publications/orca/brochures/cave-tour-map.pdf). This involved converting the surveyed passage outlines into distinct, accurately positioned digital surfaces representing the cave's roof and floor profiles. Thus, capturing its characteristic passages, chambers, and irregular conduit cross-sections. Both analogs serve as geometrically faithful karstification model to investigate geohazards associated with unpredictable karst voids.

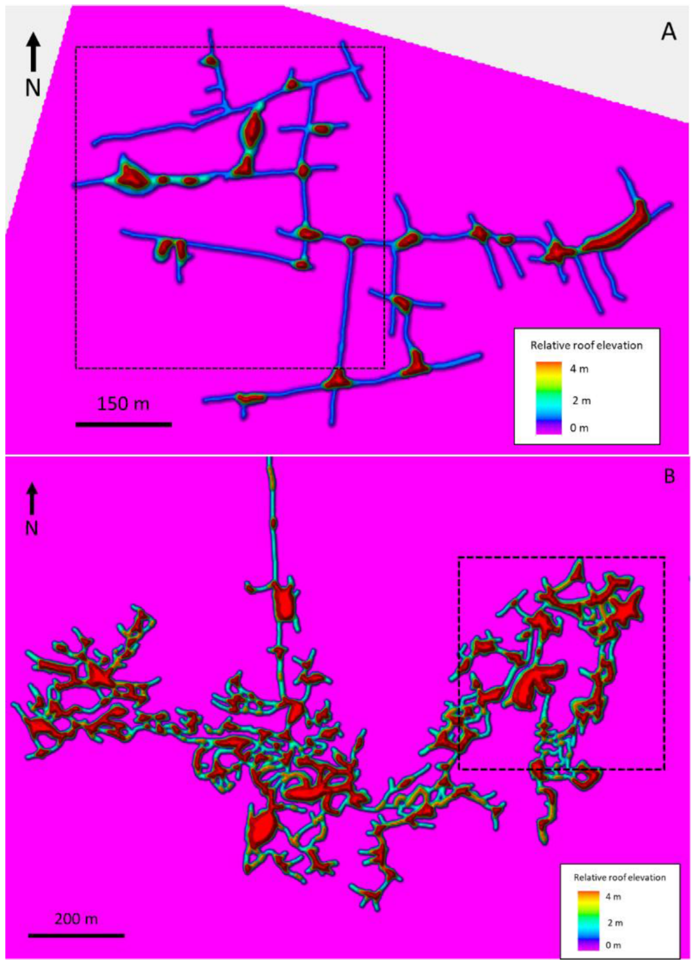

Figure 1—Digitized karst geometry represented as surface for physical modeling and numerical simulation: (a) Simple karst model analog of the Arabian fracture-induced karst system. (b) Complex karst model analog of a mature epigenic cave system inspired by Watson Cave, Oregon. Dashed polygons represent the extent of the 3D digital seismic property model.

Digital model building

A high-resolution digital seismic property models were constructed from the karst geometries for numerical seismic wave propagation simulation. The detailed spelunker-derived surfaces representing the roof and floor of both the simple (Arabian fracture-induced) and complex (Watson Cave-inspired) epigenic karst systems were sampled into a fine-scale 500m × 500m × 500m 3D geocellular grid. This grid embedded the karst voids within a background stratigraphy comprising three primary layers: a surficial sand unit overlying two distinct limestone layers. The critical bed contact between Limestone Layer 1 and Limestone Layer 2 was explicitly positioned within the grid to lie stratigraphically between the karst floor and roof surfaces. The thickness of the surficial sand layer was deliberately designed to vary inversely with the depth to the major karst chambers, resulting in increased sand thickness directly overlying the largest subsurface voids. Seismic P-wave velocity (Vp), S-wave velocity (Vs), and density for the sand, both limestone layers, and the air-filled karst voids were assigned according to the representative values detailed in Table 1, completing the property model (Figure 2).

Table 1—P-wave velocity, S-wave velocity, and density of the geological model.

	VP [m/s]	VS [m/s]	Density [g/cm³]
Sand	800	420	1.45
Layer-1	3000	1800	2.2
Layer-2	3800	2200	2.32
Karst filling	330	0	1.3

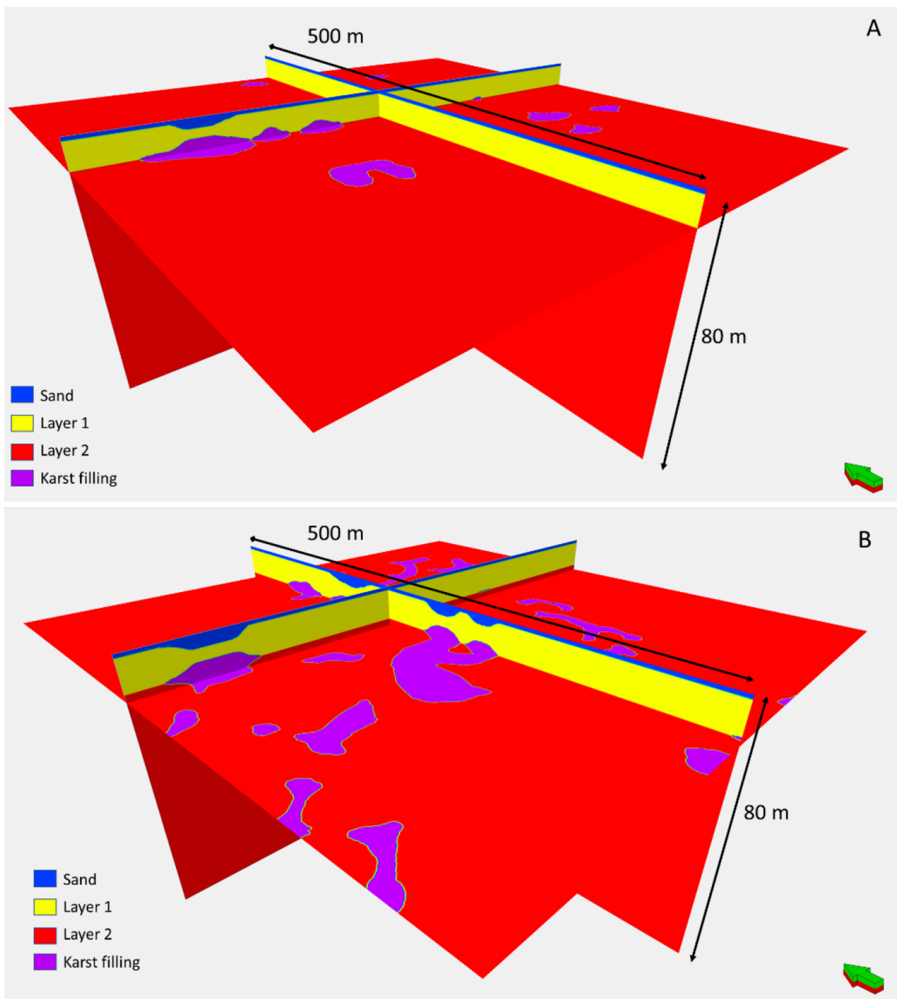

Figure 2—Example display of 3D geological models constructed from the karst geometries: (a) Simple karst model. (b) Complex karst model.

Numerical simulation

To validate the accuracy of the constructed models in replicating complex wavefields akin to those observed in field settings, we conducted 2D elastic modeling simulations. The shot was positioned at the center of the model, specifically at coordinates (130, 250), while receivers were placed at 1-meter intervals along the line Y=250. A Ricker wavelet with a central frequency of 60Hz was utilized to generate the seismic signal. The results of the numerical simulation for both simple and complex karsting models are presented in Figure 3A and 3B, respectively. A visual examination of the simulated data reveals significant phase and amplitude distortions over areas associated with shallow karsting, which is consistent with the findings from field data analysis. This congruence between simulated and real-world data suggests that the constructed models effectively capture the complex seismic behavior characteristic of karstified regions.

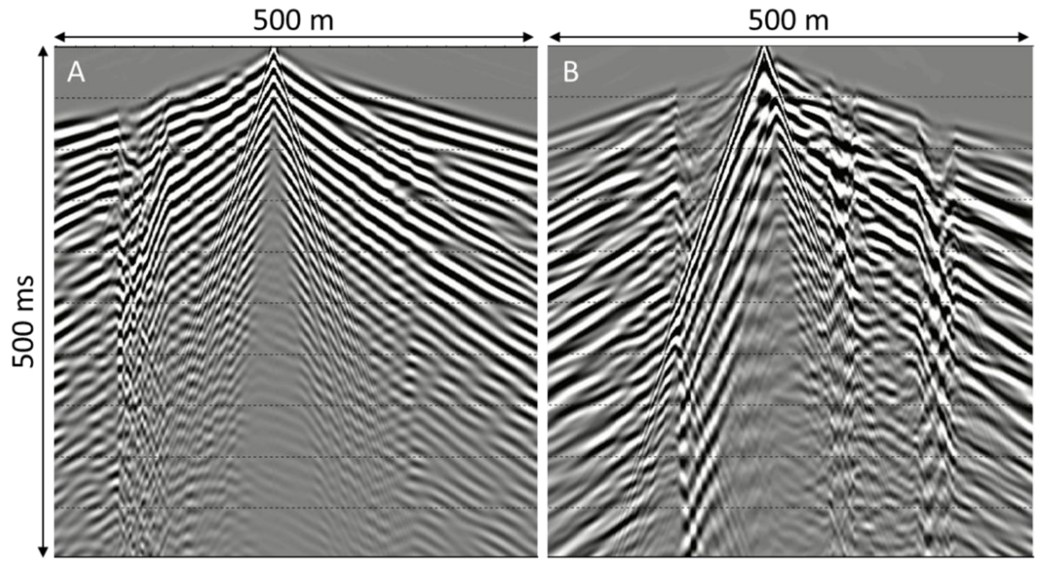

Figure 3—Example of the 2D elastic modeling for the Simple karst model (a) and Complex karst model (b).

Physical model building

Even the most advanced numerical simulation methods are based on certain assumptions and simplifications and cannot fully describe complex visco-elastic wave propagation through the kastified strata. One way to study this in a controlled environment is to perform physical modeling when the model size, properties and the data acquisition parameters are scaled appropriately. In our experiment, we downscaled our digital 3D model to a 0.5x0.5x0.5m physical prototype (scale 1:1000). We started building our model by 3D printing with the acetone solvable plastic of the surfaces corresponding to the top and bottom of the karsting (Figure 4).

Figure 4—The 3D-printed plastic model of the karst cavity system. Each void was printed separately (left) and then merged together (right).

Next, the 3D printed voids were attached to the bottom of the casting mold using pumping tubes, through which a solvent was pumped after the mold solidified in order to dissolve the plastic and form cavities. (Figure 5)

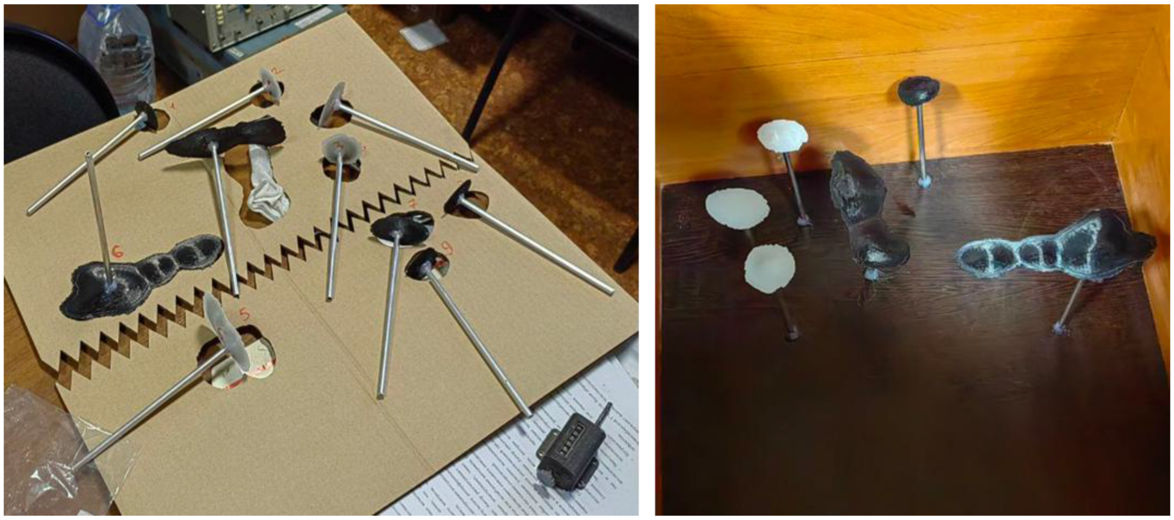

Figure 5—The process of assembling plastic void models into the casting mold. 3D printed top and bottom of each karst were attached to the solvent pumping tubes (left) and installed in the casting mold at appropriate depth.

Finally, the model was filled with cement in multiple stages to avoid disrupting the spatial arrangement of the voids (Figure 6). A carbonate-based sand was chosen to simulate a carbonate rock with ~90% calcite content. The sand was sifted to a size of 0.15-0.3 mm and then mixed with cement at a 1:1 ratio to form a concrete. The concrete hardening process was monitored by periodically measuring the seismic velocities of the block. After 20 days, the velocities had stabilized at around 4,500 m/s, consistent with typical carbonate-rock velocities. After the model solidified, an acetone solution was pumped under pressure through the pumping channels, dissolving and flushing out the plastic. Since the data measurements were taken from above using a piezoelectric sensor, the remaining conductive channels do not affect the measurements. When the concrete model was fully ready, a layer of sand, sifted to 0.25-0.25 mm, was placed above the concrete block.

Figure 6—The process of filling the model with concrete.

Passive seismic measurements

Upon completion of the physical models and full solidification of the concrete, we conducted a series of preliminary experiments to measure the standing wave resonant frequencies, as described in previous studies by Fedin et al. (2021) and Silvestrov et al. (2024). The experimental setup is illustrated in Figure 7, which depicts the configuration of the equipment used to generate and record the acoustic noise field. Two electrodynamic loudspeakers were positioned on opposite sides of the model, producing separate electrical signals in a white noise mode to generate a continuous acoustic noise field. The recording of the noise signals was performed at a sampling rate of 1 MHz using broadband piezoceramic piston-type sensors with a diameter of 0.2 cm. The sensors' axes of maximum sensitivity were carefully oriented perpendicular to the surface of the model, allowing for the primary detection of vertical vibrations. This experimental setup enabled the quick and efficient collection of high-quality data, which was subsequently analyzed to determine the standing wave resonant frequencies.

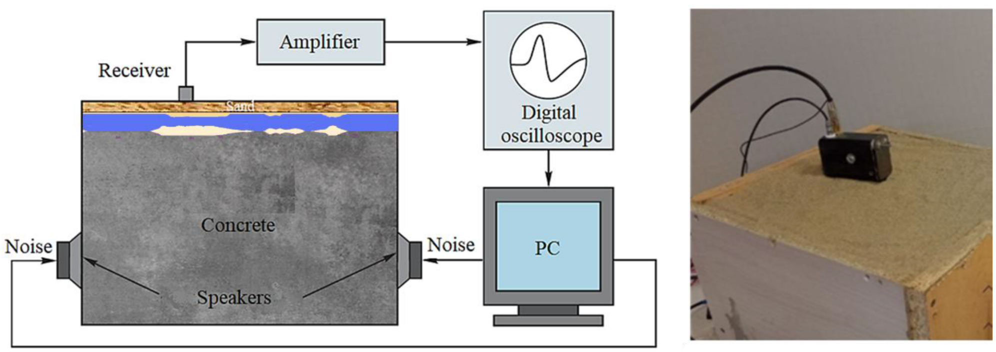

Figure 7—Experimental setup: schematic overview (left) and actual implementation in the laboratory (right).

The measurements were conducted at two distinct locations: directly above the void and over the solid rock. An exemplary amplitude spectrum of the acoustic noise signal is presented in Figure 8, illustrating the notable differences between the two recording locations. The amplitude spectrum of the signal recorded over the solid rock, denoted by the blue line, exhibits a relatively flat profile, whereas the spectrum of the signal recorded over the void displays prominent spikes. These spikes correspond to the resonant frequency of approximately 145 Hz and its harmonics, which agrees with the theoretical prediction based on the formula: $f_n = nVp/2h$, where n is the standing wave mode number, Vp is the P-velocity and h is the thickness of the layer above the void.

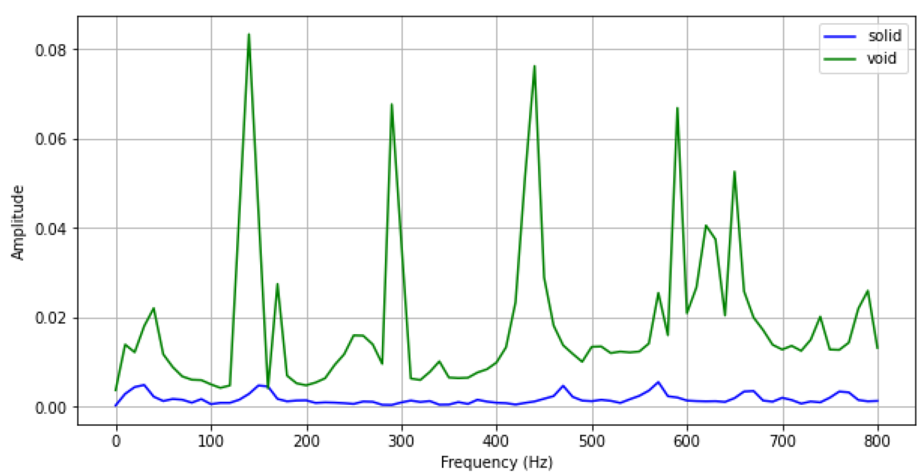

Figure 8—Amplitude spectra of the acoustic signals recorded over the void (green) an over the solid rock (blue).

Subsequent measurements were conducted at 1-centimeter intervals along the line corresponding to profile AB, as illustrated in Figure 9a. The results are presented in Figure 9b, where the fundamental mode of the standing wave resonant frequency is plotted in blue at each location, alongside the corresponding slowness values at a depth of 2cm (equivalent to 20m in the upscaled model). A remarkable correlation is observed between the measured values and the model properties, demonstrating the efficacy and applicability of the selected method for detecting shallow karsting features with high accuracy and precision.

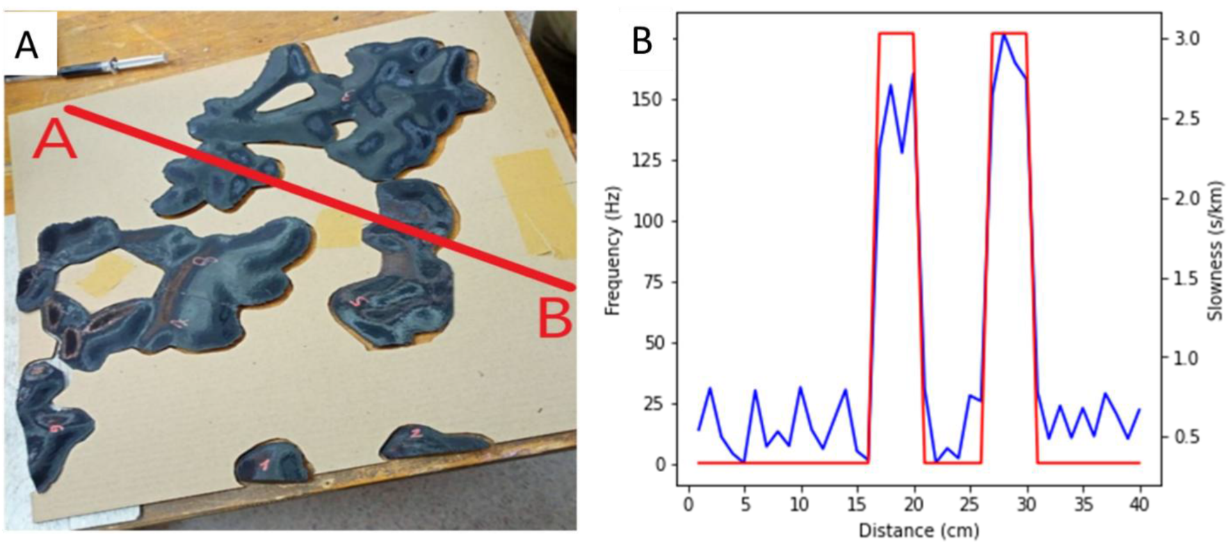

Figure 9—Standing waves resonance profiling across the model along the line AB (a) and the resulting plot of the first mode of the resonant frequency (blue) vs slowness at depth 2cm (red) along the measured profile.

Conclusions

This study introduces a pioneering interdisciplinary approach that integrates 3D modeling, additive manufacturing, and advanced laboratory experimentation to provide invaluable insights into the complex phenomena of shallow karsting development in arid environments. By combining literature examples, field observations, and laboratory measurements, we demonstrated a data-driven approach to digital model building, resulting in geologically sensible models that strike a balance between simplicity and realism. These models are sufficiently simple to be replicated as laboratory mockups, yet realistic enough to significantly impact data quality.

The application of additive manufacturing technologies enabled us to recreate the complex topology of cave systems within laboratory concrete models, allowing for the precise replication of intricate geological features. We successfully recreated both digital models as 1:1000 scaled mockups, which were subsequently utilized for laboratory measurements. The use of these physical models facilitated the collection of high-quality data, which is essential for advancing our understanding of shallow karsting phenomena.

Our research also highlights the effectiveness of passive seismic measurements and standing wave resonance analysis as a tool for detecting shallow karsting. By conducting experiments on our physical models, we demonstrated that this method can be a rapid and cost-effective means of identifying drilling hazards, thereby mitigating the risks associated with karsting. The findings of this study have significant implications for the development of more accurate and efficient methods for detecting shallow karsting, which is crucial for ensuring the safety and success of drilling operations in arid environments.

The integration of these innovative techniques has far-reaching potential for advancing our understanding of complex geological systems and addressing real-world challenges. By leveraging the strengths of each approach, we can develop more comprehensive and accurate models of shallow karsting, ultimately leading to improved drilling practices and reduced environmental risks. Furthermore, this research demonstrates the value of interdisciplinary collaboration, highlighting the benefits of combining expertise from fields such as geology, engineering, and materials science to tackle complex problems. Overall, this study showcases the potential of innovative, interdisciplinary approaches to drive breakthroughs in our understanding of the Earth's subsurface and develop effective solutions for real-world challenges.

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